Project: Worked examples
Title: Pipe pressure drop and pump duty
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Friction and fitting losses through series pipe segments at a given flow, the static head, the system curve, the duty point against a pump curve entered as points, the affinity-law shift for a speed or impeller change, and the NPSH available at the pump suction.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Design flow from the heat carried, when a heat load is given in place of a flow | $Q = \Phi/(\rho\, c_p\, \Delta T)$ |
| Mean velocity in the pipe | $v = Q/(\pi D^2/4)$ |
| Reynolds number, which decides laminar or turbulent flow | $Re = \rho v D/\mu$ |
| Friction factor for laminar flow (Re < 2300) | $f = 64/Re$ |
| Friction factor for turbulent flow, Colebrook–White by fixed-point iteration (at most 100 steps) from the Swamee–Jain value, which also cross-checks it | $1/\sqrt{f} = -2\log_{10}(\varepsilon/3.7D + 2.51/(Re\sqrt{f}))$ |
| Head lost in the pipe and its fittings | $h_f = f\,(L + L_{eq})/D \cdot v^2/2g + K\, v^2/2g$ |
| System curve, with the friction of a turbulent or transitional segment and every fitting loss scaled from the design flow as $Q^2$, and the friction of a laminar segment (Re < 2300, $h_f \propto Q$) as Q (a closed circuit has $\Sigma dz = 0$, the rise in the flow being recovered in the return) | $H_{sys}(Q) = \Sigma dz + h_{out} + \Sigma h_{f,lam}\cdot(Q/Q_o) + \Sigma h_{f,other}\cdot(Q/Q_o)^2$ |
| Water held in the pipes | $V_{pipe} = \Sigma\, \pi/4 \cdot D^2 \cdot L$ |
| Duty point, where the pump and system curves cross (every crossing is found by a scan and refined by bisection; the stable one, where the pump curve falls through the system curve, nearest the design flow is reported) | $H_{pump}(Q) = H_{sys}(Q)$ |
| Pump affinity laws for a change of speed n | $Q \propto n$; $H \propto n^2$. For a change of impeller diameter D, by the choice: a trimmed impeller of the same casing, $Q \propto D$ and $H \propto D^2$ (the default, the usual practice for a trim); or a geometrically similar pump scaled in all dimensions, $Q \propto D^3$ and $H \propto D^2$. Speed and diameter combine as $Q \propto n\cdot D\text{^}a$, $H \propto n^2\cdot D^2$, a = 1 or 3 |

| WHAT IT CALCULATES | RELATION |
|---|---|
| Net positive suction head available, to compare with what the pump requires | $NPSH_a = p_{atm}/\rho g + z_s - h_{f,suction} - p_v/\rho g$, with $p_v$ from the water correlation or, for a custom fluid, from the vapour pressure entered. The friction $h_{f,suction}$ is that of the segments marked 'suction' only, at the larger of the design flow and the duty flow the pump runs at |
| Shaft torque for the motor | $T = P/\omega$ with P the larger of the design-flow shaft power and the duty-point shaft power, at the speed $n_{pump}$; when a speed or diameter ratio shifts the duty, the shifted duty power at the shifted speed $n_{pump} \cdot n_2/n_1$ |

Basis: Flow of fluids through valves, fittings and pipe[1]; Internal flow: friction factor, minor losses, pump duty[2](Ch. 8); Flow of fluids in pipes and ducts (principles)[3](Section 4); Density of liquid water[4].

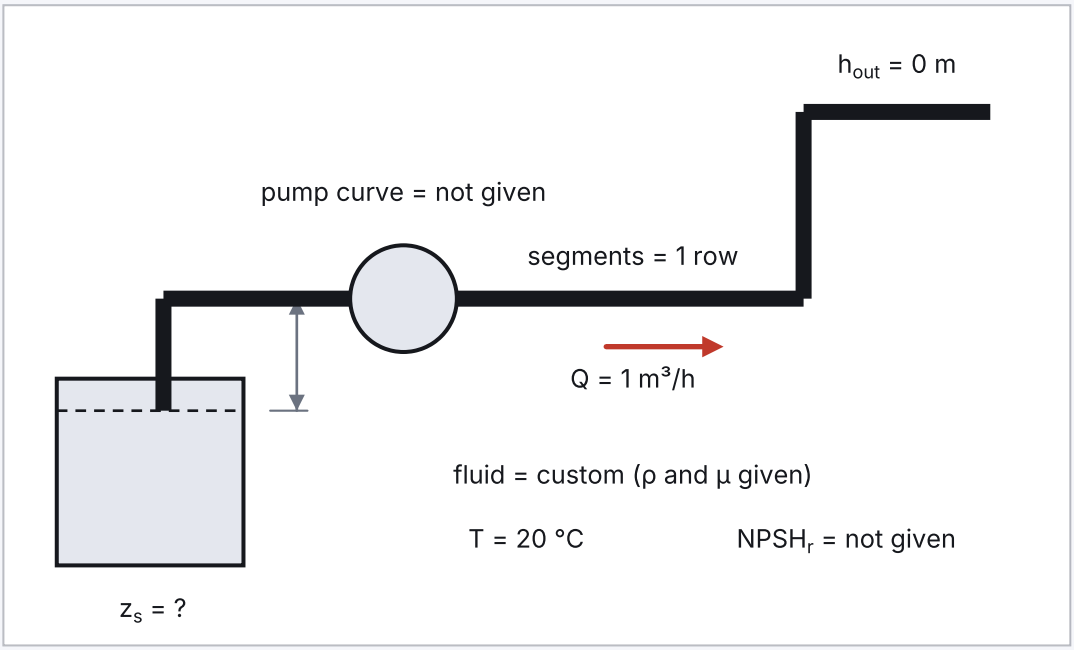



Figure 1 — A pump lifting from a suction level through the pipe segments as listed (three drawn) to an outlet that needs a pressure head. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| fluid | Fluid | custom (ρ and μ given) | | spec: Oil datasheet ISO VG 680 at 20 °C | any |
| T | Fluid temperature (mean; link B6 $t_m$ for a heating loop) | 20 | °C | assumption: Ambient 20 °C. | 0 … 100 °C |
| $\rho_{custom}$ | Density (custom fluid) | 888 | kg/m³ | spec: Oil datasheet ISO VG 680 | 500 … 2000 kg/m³ |
| $\mu_{custom}$ | Dynamic viscosity (custom fluid) | 800 | mPa·s | spec: Oil datasheet ISO VG 680 at 20 °C | 0.1 … 1000 mPa·s |
| Q | Design flow rate (blank when a heat load gives it) | 1 | m³/h | spec: Transfer pump duty | 0.01 … 100000 m³/h |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| segments | Pipe segments in series | 1 row(s) | | drawing: DRG-L-020 rev A | any |
| circuit | Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored) | open (the pump lifts the water by the rises entered) | | default | any |
| $h_{out}$ | Pressure head required at the outlet | 0 | m | assumption: Open discharge. | 0 … 1000 m |
| $v_{max}$ | Velocity limit (warning) | 3 | m/s | assumption: Practice limit. | 0.5 … 10 m/s |
| $n_{ratio}$ | Speed ratio $n_2/n_1$ (affinity) | 1 | | default | 0.2 … 2 |
| $D_{ratio}$ | Impeller diameter ratio $D_2/D_1$ (affinity; the law is chosen below) | 1 | | default | 0.7 … 1.1 |
| $D_{law}$ | Law for the diameter ratio: a trimmed impeller of the same casing ($Q \propto D$, $H \propto D^2$) or a geometrically similar pump ($Q \propto D^3$, $H \propto D^2$) | impeller trim ($Q \propto n \cdot D$, $H \propto n^2 \cdot D^2$) | | default | any |
| $\eta_{pump}$ | Pump efficiency at duty (for shaft power) | 0.7 | | default | 0.2 … 0.95 |
| $p_{atm}$ | Atmospheric pressure (NPSH) | 101.325 | kPa | default | 50 … 110 kPa |
| $z_s$ | Static head at the pump suction (+ flooded, − lift) | 0 | m | default | -10 … 100 m |

| TABLE ROW | VALUES |
|---|---|
| segments[1] | name = transfer line; L = 40 m; D = 50 mm; eps = 0 mm; K = 0; L_eq = 0 m; dz = 0 m; side = discharge |

## 3. Assumptions

A1. Steady, incompressible, fully developed flow; all segments carry the full design flow in series. (Darcy–Weisbach.)
A2. Fitting losses scale with velocity head (K method) or add pipe length (equivalent-length method); both may be used on one segment. (Crane TP-410 practice.)
A3. The system curve scales the friction of turbulent and transitional segments and all fitting losses with $Q^2$ from the design flow (the friction factor is held at its design-flow value); the friction of a laminar segment scales with Q; the static part is independent of flow. ($H_{sys} = H_{static} + k_1 Q + k_2 Q^2$.)
A4. Water properties are from published correlations at atmospheric pressure (0–100 °C); any other fluid is entered with its own density and viscosity and their provenance. (Kell 1975; Vogel fit; Hyland & Wexler 1983.)
A5. T (Fluid temperature (mean; link B6 $t_m$ for a heating loop)) is taken as 20 °C. (Ambient 20 °C.)
A6. $h_{out}$ (Pressure head required at the outlet) is taken as 0 m. (Open discharge.)
A7. $v_{max}$ (Velocity limit (warning)) is taken as 3 m/s. (Practice limit.)

## 4. Working

[5] $\rho = \rho_{custom}$
$= 888\ kg/m^3$
$= 888\ kg/m^3$
Density (as entered)

[6] $\mu = \mu_{custom}$
$= 800\ mPa \cdot s$
$= 800\ mPa \cdot s$
Dynamic viscosity (as entered)

[7] $T_{line}$ = T (no $T_{line}$ entered)
$= 20\ °C$
$= 20\ degC$
Design temperature of the line to insulate: the fluid temperature T, since no $T_{line}$ was entered

[11] $Q_{design} = Q$
$= 1\ m^3/h$
$= 1\ m^3/h$
Design flow (as entered)

[12] $v_{s1}$ = transfer line: $Q / (\pi D^2/4)$
$= 1\ m^3/h / (\pi \times 50^2\ mm^2 / 4)$
$= 0.14147\ m/s$
Velocity, transfer line

[13] $Re_{s1}$ = transfer line: $\rho\ v\ D / \mu$
$= 888 \times 0.14147 \times 0.05 / 0.8$
$= 7.8516$
Reynolds number, transfer line

[14] $f_{s1}$ = transfer line: Darcy friction factor
= 64 / Re (laminar)
$= 8.1512$
Friction factor, transfer line

[15] $h_{f,s1}$ = transfer line: $f (L + L_{eq})/D \cdot v^2/2g + K\ v^2/2g$
$= 8.1512 \times (40 + 0) / 0.05 \times 0.0010204 + 0 \times 0.0010204$
$= 6.6542\ m$
Head loss, transfer line

[16] $h_{f,total} = \Sigma\ h_f$
$= 1$ segment(s)
$= 6.6542\ m$
Total friction and fitting loss at the design flow

[18] $V_{pipe} = \Sigma\ \pi/4 \cdot D^2 \cdot L$
$= 1$ segment(s)
$= 78.54\ L$
Water held in the pipe segments

[19] $H_{static} = \Sigma\ dz + h_{out}$
$= 0 + 0$
$= 0\ m$
Static head

[20] $H_{sys} = H_{static} + h_{f,total}$
$= 0\ m + 6.6542\ m$
$= 6.6542\ m$
System head at the design flow

[21] $dp_{total} = \rho * 9.80665\ m/s^2 * h_{f,total}$
$= 888\ kg/m^3 \times 9.80665\ m/s^2 \times 6.6542\ m$
$= 57.947\ kPa$
Friction pressure drop at the design flow

[22] $dp_{sys}$ = ρ * 9.80665 m/s^2 * $H_{sys}$
= 888 kg/m³ × 9.80665 m/s^2 × 6.6542 m
= 57.947 kPa
System pressure at the design flow
[23] $P_{hyd}$ = ρ * 9.80665 m/s^2 * $Q_{design}$ * $H_{sys}$
= 888 kg/m³ × 9.80665 m/s^2 × 1 m³/h × 6.6542 m
= 0.016096 kW
Hydraulic power at the design point
[26] $v_{peak}$ = highest segment velocity
= 0.14147 m/s
= 0.14147 m/s
Highest velocity
[27] $u_v$ = $v_{peak}$ / $v_{max}$
= 0.14147 / 3
= 0.047157
Velocity utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $Q_{design}$ | 1 m³/h | Design flow |
| $H_{static}$ | 0 m | Static head |
| $V_{pipe}$ | 78.54 L | Water held in the pipes |
| $H_{sys}$ | 6.654 m | System head at Q |
| $h_{f,total}$ | 6.654 m | Σ $h_f$ |
| $dp_{total}$ | 57.95 kPa | Friction Δp |
| $dp_{sys}$ | 57.95 kPa | System Δp |
| $P_{hyd}$ | 0.01610 kW | Hydraulic power |
| $v_{peak}$ | 0.1415 m/s | Highest velocity |
| $T_{line}$ | 20 °C | Line design temperature |
| $v_{s1}$ | 0.1415 m/s | Velocity, transfer line |
| $Re_{s1}$ | 7.8516 | Re, transfer line |
| $f_{s1}$ | 8.1512 | f, transfer line |
| $h_{f,s1}$ | 6.654 m | $h_f$, transfer line |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Velocity ≤ limit | $u_v$ = 0.05 | $u_v$ ≤ 1 | 0.05 | |

## 7. Applicability

WARNING segments: transfer line is laminar (Re = 7.8516); f = 64/Re. The system curve and the duty point scale its friction with Q ($h_f$ ∝ Q), not Q². [laminar]

Defaults used (inputs left empty):
circuit: not entered, so the default of 'open (the pump lifts the water by the rises entered)' is used.
$n_{ratio}$: not entered, so the default of 1 is used.
$D_{ratio}$: not entered, so the default of 1 is used.
$D_{law}$: not entered, so the default of 'impeller trim (Q ∝ n·D, H ∝ n²·D²)' is used.

$\eta_{pump}$: not entered, so the default of 0.7 is used.
$p_{atm}$: not entered, so the default of 101.3 kPa is used.
$z_s$: not entered, so the default of 0 m is used.

## 8. Verification

```
Template bs.pipe-pump@1.9.0 Content hash 15a0bf02d74b6e79 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References

1. Crane Co. Flow of Fluids Through Valves, Fittings and Pipe. Technical Paper No. 410. Crane Co.
2. Çengel YA, Cimbala JM. Fluid Mechanics: Fundamentals and Applications. 4th ed. McGraw-Hill Education; 2018.
3. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.
4. Kell GS. Density, thermal expansivity, and compressibility of liquid water from 0° to 150°C: correlations and tables for atmospheric pressure and saturation reviewed and expressed on 1968 temperature scale. J Chem Eng Data. 1975;20(1):97-105.